FLEXIPATH

PREDICTIVE · TRAFFIC-AWARE CONNECTIVITY ENGINE

ORGANISED BY
ADVANCED TECHNOLOGY RESEARCH COUNCIL
ADVANCED TECHNOLOGY RESEARCH COUNCIL

SWITCH BEFORE THE LINK FAILS.

Keeps a drone's **control link**, voice and video alive as it moves between **Wi-Fi, 5G, satellite and wired** networks, by predicting the failure and moving before it happens.



TEAM
STAND 27 · PROJECT 5007

FlexiPath
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PROBLEM STATEMENT · EDGE GROUP CHALLENGE STATEMENT

“How might we keep a user's experience unbroken as they move between Wi-Fi, mobile, satellite and wired networks?”

Maintaining a seamless connection as users move between network technologies and coverage areas is one of the greatest challenges in modern communication. Interruptions degrade the user experience and hit users' critical applications. — EDGE Group, ATP 2026

WHY IT MATTERS · OUR SCENARIO

Emergency response, industrial automation, connected transport and defence cannot wait for a reconnect. We chose a **drone beyond its pilot's sight**: in our model its command link may go silent for at most **300 ms** before it must stop and hover.

1. **DETECT** — Measure signal, loss, delay and jitter ten times a second.
2. **PREDICT** — Extend the trend; find the time to failure. The route map warns 8 s ahead.
3. **PREPARE** — Wake the best spare radio and test it with tiny heartbeats.
4. **REPLICATE** — Copy only critical traffic on both paths for 400 ms.
5. **SWITCH** — Make the tested path primary. Zero gap on predicted moves.
6. **CONTINUE** — Old radio back to sleep; every session keeps one address.

PROPOSED SOLUTION

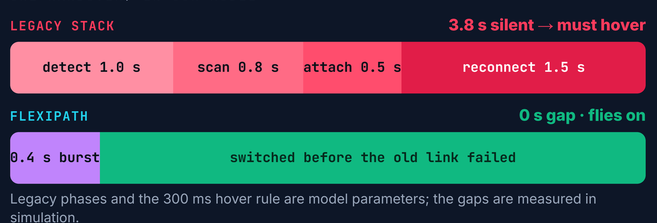
Predict, prepare, then switch

Today's devices wait for a network to fail before they move to the next one. In our model, that break-before-make switch leaves a legacy stack silent for **3.8 seconds**. A drone whose control link is silent for more than 300 ms must stop and hover, so every handover becomes a risk.

FlexiPath is a predictive, traffic-aware connectivity engine that sits under the applications. It watches every link ten times a second and spots a fading signal seconds before it fails. It wakes the next radio and tests it, sends only the critical commands over both networks for 0.4 seconds, then switches with no gap. A fixed anchor relay keeps every session open, so the ground station always sees one address, whatever radio the drone is using. Spare radios sleep the rest of the time to save power.

We built it as a working digital twin: two identical drones fly the same mission through Wi-Fi, 5G and satellite coverage, one with FlexiPath and one with a legacy stack, under the same radio conditions. A deterministic simulation and a headless benchmark measure both against two existing mechanisms.

ONE HANDOVER, IN OUR MODEL



Legacy phases and the 300 ms hover rule are model parameters; the gaps are measured in simulation.

TECHNICAL EXPLANATION

Six stages, ten times a second

The engine runs a six-stage loop ten times a second. **DETECT** measures signal, loss, delay and jitter. **PREDICT** fits a 3-second trend and estimates the time to failure, the moment loss would reach 15 %; a coverage map from earlier flights warns 8 seconds before a known dead zone. **PREPARE** wakes the best spare modem from sleep (1.2 W to 1.8 W) and checks it with 64-byte heartbeats every 100 ms. **REPLICATE** sends critical traffic on both paths for a 400 ms burst (3.4 W); voice joins only if its loss exceeds budget, and bulk uploads are never copied. **SWITCH** makes the tested path primary with a 0 ms gap, and **CONTINUE** puts the old radio back to sleep.

The receiver keeps the first copy of each sequence number and drops the late duplicate, so there is no head-of-line delay. All sessions ride a UDP tunnel to an anchor relay, so the remote end never sees an address change and no connection resets. If an outage comes with no warning, failover takes 160 ms in our model. Every decision is recorded with the values that caused it, so the dashboard can explain it.



The 400 ms burst in the twin: purple beams carry the control link over Wi-Fi and 5G at once.

SOLUTION FEATURES

What FlexiPath does differently

- **Predicts instead of reacting.** It acts 1.2–4.6 s before the path fails; reactive multipath acts 0.1–0.5 s after it (scenarios A–D).
- **Traffic-aware.** Three classes: control, voice and bulk. Only the critical class is ever duplicated, about 1–3 % of protected bytes.
- **Power-aware.** Spare radios sleep in DRX. FlexiPath draws 1.25–1.39 W, about 8 % more than a single radio, against 2.65–3.39 W for always-on duplication.
- **Route foresight.** A coverage map from earlier flights warns of dead zones 8 s ahead. In the urban-canyon scenario FlexiPath kept the control link on 20 of 20 flights; reactive multipath on 0 and always duplicate on 8.
- **One address for the whole flight.** Zero TCP resets and zero visible IP changes in the predicted handovers.
- **Explainable.** A decision card shows the current path, trend, time to failure, alternate score, radio tier, action and reason.
- **Measured, not asserted.** 160 paired simulated flights (8 scenarios × 5 seeds × 4 speeds) with 95 % confidence intervals, ablations, held-out seeds and sensitivity runs. One command reproduces every number.
- **Honest baselines.** A legacy single-path stack, reactive multipath (MPTCP backup subflow / 3GPP ATSSS active-standby) and always duplicate (MPTCP redundant scheduler / PDCP duplication), all on the same seeds.
- **Live prototype.** An interactive 3D digital twin with a guided demo runs in any browser at flexipath.vercel.app.
- **Application-neutral.** Designed to sit under any command-and-control application; the application itself does not change.

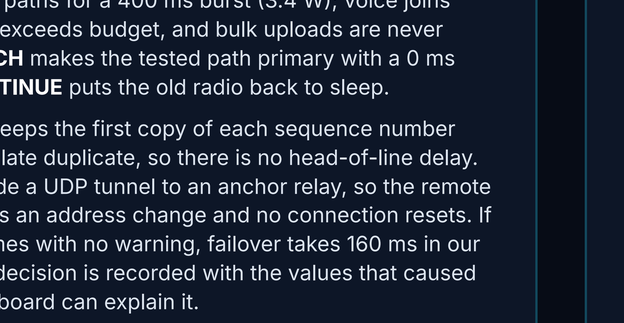
100 %
of 160 simulated flights kept the control link with FlexiPath · **legacy 0 %** · reactive multipath fails in 3 of 8 scenarios
MEASURED · SIM · OUR 300 ms RULE

35 %
fewer critical packets lost than reactive multipath (about 20 % with equal outage detection); holds on held-out seeds
MEASURED · SIM

2.1–2.6×
less radio power than always-on duplication, for nearly the same delivery (**99 % vs 99–100 %**)
MEASURED · SIM

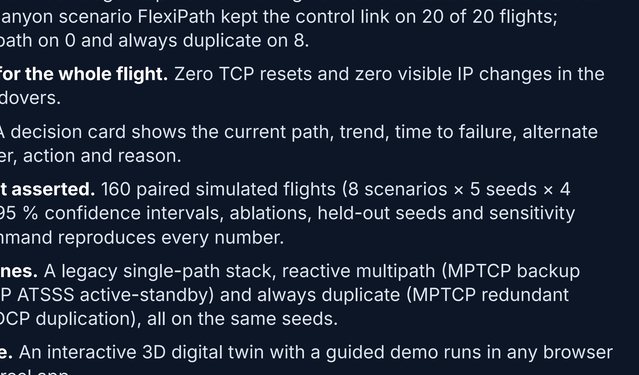
4–7 pp/W
reliability gained per extra watt over reactive multipath; always duplicate adds only 0.3–0.9 pp/W
MEASURED · SIM

Reliability vs radio power
MEASURED · SIM
Critical (C2) packets delivered against mean radio power · scenarios A–D, G, H · each marker is the mean of 20 paired flights



Critical-packet delivery against mean radio power for the three multipath stacks, scenarios A–D, G and H. Each marker is the mean of 20 paired flights; FlexiPath sits near duplication's reliability at less than half its power.

LIVE PROTOTYPE · TRY IT AT THE BOOTH



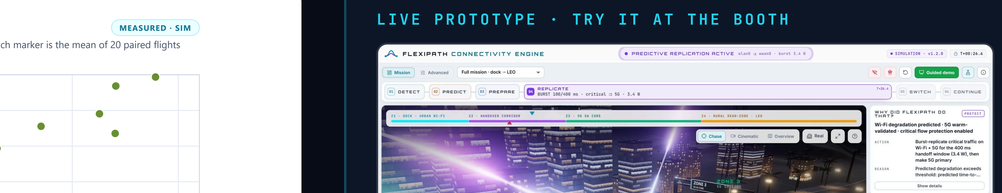
Guided demo & simulator
flexipath.vercel.app

2-minute film
flexipath.vercel.app/video

FUTURE IMPROVEMENTS & REAL-LIFE PLAN

- **Real radios.** Port the engine to a Linux companion computer with Wi-Fi, 5G and satellite modems, and measure real handovers with our probe script. Today every number is simulated.
- **Fairer comparison.** Give the reactive baseline the same 60 ms outage detection as FlexiPath (our lead in lost packets then falls from about 35 % to about 20 %) and add signal-threshold triggers such as 3GPP measurement events.
- **Better prediction.** Replace the linear trend with a model trained on flight logs, and let a fleet share its coverage maps so every flight learns from the last.
- **Closer radio model.** Air-to-ground propagation, antenna tilt and altitude effects, shared capacity between traffic classes, and real LEO constellation timing.
- **Security and standards.** Encrypt the anchor tunnel and align it with 3GPP ATSSS and MPTCP so it can run beside existing stacks.
- **Real-life plan.** A 3–6 month pilot with one drone operator, then industrial sites and emergency response, where a silent control link is not an option.

SYSTEM DESIGN · ONE ENGINE, SEVERAL RADIOS, CONTINUOUS SESSIONS



DESIGNED TO SIT UNDER ANY COMMAND-AND-CONTROL APPLICATION · every decision is logged with its reason

Simulation prototype. Every number on this poster is a simulated measurement or a model parameter; we claim no live-network measurements. Baselines are modelled on RFC 8684 (MPTCP), 3GPP TS 23.501 (ATSSS) and 3GPP TS 38.323 (PDCP duplication); the hover budgets follow 3GPP TS 22.125.

AI-use declaration (ATP 2026 FAQ 09): Anthropic Claude assisted with the code, documentation, deck, video and this poster; the video narration is an AI voice. The concept, the design choices and the review of results are the team's own.

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